PAUL GIBSON

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EDUCATION

Georgia Institute of Technology — Atlanta, GA — Expected May 2028

Bachelor of Science in Industrial and Systems Engineering — **GPA: 3.55 / 4.00**

Relevant Coursework: Engineering Optimization (LP/MIP, Gurobi), Stochastic Manufacturing & Service Systems (Queueing & Inventory Control), Regression & Forecasting, Database Systems (Relational Schemas, SQL), Probability & Statistics, Linear Algebra.

TECHNICAL SKILLS

- **Languages & Optimization:** Python (pandas, NumPy, scikit-learn, gurobipy), SQL (MySQL), Gurobi Optimizer, R (foundations)
- **Data Analytics & Tools:** MS Excel (Pivot Tables, Advanced Formulas, Data Modeling), Git/GitHub, Matplotlib, Seaborn, REST APIs
- **Operations Focus:** Supply Chain Optimization, Demand Forecasting, Warehouse Logistics, Inventory Control, Material Handling

PROFESSIONAL EXPERIENCE

Second Harvest of South Georgia — Valdosta, GA — June 2025 – August 2025

Warehouse Operations Intern

- Fulfilled and staged up to **15 bulk palletized orders daily (~10+ tons of freight)** across the facility using electric pallet jacks.
- Managed partner agency intake, categorizing stock by cost tiers, conducting bulk weigh-ins, and logging web portal transactions.
- Conducted routine cycle counts across hundreds of SKUs, identifying inventory variances and reconciling physical stock against digital logs to maintain high record accuracy.
- Coordinated with warehouse leads to audit pallet manifests and stage outgoing freight at dock doors, ensuring accurate and timely agency order fulfillment.

TECHNICAL PROJECTS

Healthcare Capacity & Influenza Admissions Nowcasting Pipeline — Fall 2026

ISyE 4031: Regression & Forecasting — Python (NumPy, pandas, Matplotlib), OLS, CDC NHSN & NOAA Data

- Engineered an end-to-end forecasting pipeline analyzing **210 weeks** of CDC hospital respiratory admissions and NOAA weather data to predict weekly Georgia hospital bed demand from temperature shifts.
- Enforced temporal cutoff quarantines and **FluSight Git version control** to eliminate data leakage and track public health reporting revisions across retrospective and prospective horizons.
- Evaluated forecast calibration using **50%, 80%, and 95% Student-*t* prediction intervals**, validating model reliability through residual diagnostics, seasonal heteroskedasticity analysis, and Cook's distance sensitivity refits.

ACC Football Conference Operations Relational Database — Fall 2026

CS 4400: Introduction to Database Systems — MySQL, Relational Schema Design, EERD Modeling

- Architected a normalized **16-table relational database** from an Enhanced Entity-Relationship Diagram (EERD) to centralize ACC team rosters, stadium logistics, and game scheduling.
- Enforced **19 foreign-key referential integrity constraints** (`CASCADE`, `RESTRICT`, `SET NULL`) and subclass inheritance separating offensive and defensive player statistics.
- Designed **9 analytical SQL views** that automated conference standings, computed tiebreaker win rates, identified unscheduled matchups, and flagged personnel availability conflicts.

Multi-Plant Production, Inventory & Facility Location Optimization — Fall 2026

ISyE 3133: Engineering Optimization — Python (gurobipy), Linear & Mixed-Integer Programming (LP/MIP)

- Formulated a 4-week multi-plant, multi-product LP in **Gurobi** minimizing production, freight, inventory holding, and overtime costs (**$43.8k optimal plan**) while maintaining **100% regional demand fulfillment**.
- Conducted shadow-price (dual) bottleneck analysis on plant labor capacity, linearized a 60% production-share cap, and benchmarked Primal vs. Dual Simplex (up to **90,000 variables**) and MIP branch-and-bound cutting planes.

NBA Game Outcome & Team Metric Predictive Analysis — Spring 2026

CS 2316: Data Manipulation & Analysis — Python (pandas, scikit-learn, Matplotlib, Seaborn, nba_api)

- Built an automated data pipeline using **pandas** and **nba_api** to clean multi-season team and player statistics, training multivariate regression models in **scikit-learn** to predict team win differentials.
- Evaluated efficiency ratings, pace, and shooting percentages, generating correlation heatmaps and regression plots in **Matplotlib** and **Seaborn** to quantify the relative impact of offensive versus defensive metrics on team win percentage.

LEADERSHIP & CAMPUS INVOLVEMENT

Tau Kappa Epsilon Fraternity — Atlanta, GA — August 2025 – May 2026

Head Social Chair (Jan 2026 – May 2026) | Assistant Social Chair (Aug 2025 – Dec 2025)

- Directed chapter event operations, managing a **$60,900 operating budget** and achieving **0% budget overrun** across the term.
- Negotiated contracts with venues, security, and vendors for major functions with 200+ attendees, ensuring risk compliance.
- Coordinated collaborative mixers with partner student organizations for up to 70 attendees, overseeing scheduling and setup.